

# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE

### Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Managed **crash history records** and performed safety studies, addressing over 50 data record requests annually for the traffic engineering team.
- Conducted and reviewed **traffic engineering studies**, including highway capacity analysis and traffic volume data collection, supporting project planning and design phases.
- Reviewed and performed design of **traffic-related elements** such as Maintenance of Traffic (MOT), Signing, and Pavement Markings, ensuring compliance with the Maryland Manual on Uniform Traffic Control.
- Developed and enhanced **geographic information system (GIS) based tools** for traffic data and asset management, improving data accessibility for internal stakeholders.
- Automated the generation of **scheduled reports** on traffic volume and classification data using Python scripts, reducing manual reporting effort by approximately 15 hours per month.
- Supported production environments with strict SLAs for financial operations dashboards, contributing to **transaction-flow monitoring** and operational alerting for over 3,000 failed transactions weekly.
- Containerized Java Spring Boot services using **Docker** and deployed them to **AWS EC2** instances, managing application configurations via Jenkins pipelines for streamlined CI/CD.

## SKILLS

**Traffic Engineering** – Maintenance of Traffic, Signing, Pavement Marking, Traffic Control Devices, Highway Capacity Analysis, Traffic Studies, Crash Analysis, GIS Tools

**Software Development** – Java, Spring Boot, Python, REST APIs, SQL, Docker, Git, Jenkins

**Data Analysis & Reporting** – Power BI, Tableau, R, pandas, NumPy, DAX, Power Query, Data Modelling

**Cloud & Infrastructure** – AWS (EC2, S3, Lambda, CloudWatch), Linux, Shell Scripting

**Enterprise Systems & Support** – ERP Integration, Production Support, Incident Response, Change Management, Jira, Confluence, Splunk, Dynatrace

## PROJECTS

### Traffic Data Analysis and Reporting Platform (HCL Technologies)

- Developed a Python-based data pipeline to ingest and process traffic volume and classification data from multiple sources, handling approximately 500,000 records daily.
- Built interactive **Power BI dashboards** to visualize traffic patterns, identify congestion points, and support operational decision-making for transportation stakeholders.
- Implemented automated reporting scripts in Python that generated weekly summaries of key traffic metrics, reducing manual compilation time by 10 hours per week.

### GIS-Enhanced Traffic Asset Management Tool (HCL Technologies)

- Collaborated with operations teams to design and implement enhancements for a **geographic information system (GIS)** tool used for tracking transportation assets.
- Wrote **SQL queries** against PostgreSQL to extract and validate asset data, ensuring accuracy for over 10,000 infrastructure components.
- Developed shell scripts on Linux to automate data synchronization tasks between the GIS database and other enterprise systems, improving data consistency across platforms.

## EDUCATION

### M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

### B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

## CERTIFICATIONS

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- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)